Curriculum Vitae

Won-Chang Choi

Department of Physics and Chemistry
Daegu Gyeongbuk Institute of Science and Technology (DGIST)
Daegu 42988, Republic of Korea

Phone: +82 10-9996-9812
Email: nanady11@dgist.ac.kr
Wonchang0605@gmail.com

Education

Feb. 2015 - Feb. 2020| B.S. in School of Basic Science, DGIST

Mar. 2020 - Aug. 2026| Integrated M.S. & Ph.D. program in Department of Physics and Chemistry, DGIST

Advisor: Prof. Jung-Il Hong

Professional Summary

Experimental spintronics researcher specializing in electrical control of artificial non-collinear spin structures, antiferromagnetic exchange bias, and ferrimagnetic compensation phenomena. Experienced in thin-film growth, nanofabrication, magnetotransport, and magneto-optical characterization.

Research Interests

Spintronics, Helical spin structures, Spin-orbit torque, Exchange bias, Ferrimagnetism, Antiferromagnetism

Research Experience

R1) Formation of helical spin alignment in the AFM/FM/AFM trilayers by spin-orbit torque controlled exchange bias (*APL Materials*, 11, 121115 (2023))

- Demonstrated the formation of helical spin structures in AFM/FM/AFM trilayers using spin-orbit torque

- Achieved electrical control of chirality and investigated chirality-dependent domain-wall dynamics using Kerr microscopy

- Established the correlation between spin configuration and magnetotransport properties

R2) Electrical tuning of ferrimagnetic spin alignment angle and compensation temperature in Pt/$IrMn_3$/CoGd Multilayers (*Advanced Functional Materials*, in press, 2026)

- Demonstrated electrical tuning of the ferrimagnetic compensation temperature by up to ~110 K through spin-orbit torque manipulation of exchange coupling.

- Correlated the compensation temperature shift with electrically controlled changes in the CoGd spin configuration.

- Attributed the effect to selective exchange coupling between $IrMn_3$ and the Co sublattice.

R3) Electrical switching between uniform and 180° helical spin states (*Manuscript in preparation*)

- Realized reversible electrical switching between uniform and 180° helical spin configurations in Pt/NiO/Ni/NiO multilayers

- Demonstrated nonvolatile resistance switching driven by electrically controlled spin alignment.

- Characterized the magnetotransport properties and spin-wave dynamics of the helical spin states through electrical measurements and Brillouin light scattering.

Publications

1) **W.-C. Choi**, T.-B. Sim, K. Park, T.-H. Kim, S. Lee, B.X. Tran, S. Kim, C.-Y. You, J.-I. Hong, "Electrical tuning of ferrimagnetic spin alignment angle and compensation temperature in Pt/$IrMn_3$/CoGd multilayers", *Adv. Funct. Mater*. in press (2026)

2) **W.-C. Choi**, S. Yoon, H.-J. Kim, J.-H. Ha, K.-J. Park, E. Baek, D.-R. Kim, Y. Shin, C.-Y. You, J.-W. Kim, J.-I. Hong, "Formation of helical spin alignment in the AFM/FM/AFM trilayers by spin orbit torque controlled exchange bias", *APL Mater.* 11, 121115 (2023)

3) K. Park, J. Cho, S. Lee, J. Cho, J.-H. Ha, J. Jung, D. Kim, **W.-C. Choi**, J.-I. Hong, C.-Y. You, "Control of ferromagnetism of vanadium oxide thin films by oxidation states", *Adv. Funct. Mater.* 35, 2422966 (2025)

4) B.X. Tran, J.-H. Ha, **W.-C. Choi**, S. Yoon, T.-H. Kim, J.-I. Hong, "Field-free control and switching of perpendicular magnetization by voltage induced manipulation of RKKY interaction", *Appl. Phys. Lett*. 124, 112409 (2024)

5) C. Kim, **W.-C. Choi**, K.-Y. Moon, H.-J. Kim, K. An, B.-G. Park, H.-Y. Kim, J.-I. Hong, J. Kim, Z.Q. Qiu, Y. Kim, C. Hwang, "Dynamics of weak magnetic coupling by x-ray ferromagnetic resonance", *J. Appl. Phys*. 133, 173906 (2023)

6) B.X. Tran, S. Yang, H.-J. Kim, J.-H. Ha, S. Yoon, C. Kim, K.-W. Moon, **W.-C. Choi**, C. Hwang, J.-I. Hong, "Electrical manipulation of oxygen ion migrations for the voltage-controlled adjustment of magnetic anisotropy and optimized skyrmion generations", *Adv. Electron. Mater*. 8, 2200770 (2022)

7) H.-J. Kim, S.-G. Je, K.-W. Moon, **W.-C. Choi**, S. Yang, C. Kim, B.X. Tran, C. Hwang, J.-I. Hong, "Programmable dynamics of exchange-biased domain wall via spin-current-induced antiferromagnet switching", *Adv. Sci.* 8, 2100908 (2021)

8) H.-J. Kim, S. Yoon, J.-H. Ha, **W.-C. Choi**, J.-I. Hong, "Free tuning of exchange bias via resettable alignment of antiferromagnetic Néel axes using mechanical vibrations", *Acta Mater.* 210, 116821 (2021)

Conference Presentations

1) "Artificial helical spin structures in AFM/FM/AFM multilayers set by spin-orbit-torque control of AFM", ***MMM 2023***, **Oral presentation**, Dallas, TX, USA, Oct. 2023

2) "Electrical tuning of compensation temperature via selective exchange coupling in Pt/$IrMn_3$/CoGd multilayers", *Materials Research Society of Korea*, **Oral presentation**, Jeju, Korea, Apr. 2025

3) "Electrical tuning of spin alignment of ferrimagnet and its compensation temperature by SOT in exchange-coupled CoGd/$IrMn_3$ thin film Multilayers", The Korean Magnetic Society, **Oral presentation**, Busan, Korea, Nov. 2025

4) "Electrical tuning of the compensation temperature in HM/AFM/FIM multilayers via spin-orbit torque- controlled exchange bias", The Korean Physical Society, **Oral presentation**, Daejeon, Korea, Apr. 2026

Technical Skills

- **Thin-film growth**: magnetron sputtering, pulsed laser deposition (PLD)
- **Device fabrication**: photolithography, thermal scanning probe lithography (NanoFrazor), ion beam etching
- **Magnetic Characterization**: VSM, MPMS-SQUID, Kerr Microscopy, BLS
- **Transport & Cryogenics**: PPMS, probe station, temperature-dependent magnetotransport measurements
- **Structural Characterization**: HR-XRD, XRR, AFM, SEM/EDS
- **Software**: Python, MATLAB, Origin, LabVIEW, micromagnetic simulation (MuMax3)

References

Professor Jung-Il Hong, *Daegu Gyeongbuk Institute of Science and Technology* (DGIST), jihong@dgist.ac.kr

Professor Chun-Yeol You, *Daegu Gyeongbuk Institute of Science and Technology* (DGIST), cyyou@dgist.ac.kr